A Coherent-Packet Obstruction for Periodic Fractional Schrödinger Strichartz Estimates

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Status. Full counterexample, pending human review. For every $0 < \alpha < 1$, the regularity $\gamma_0 = (1-\alpha)/4$ proposed in Appendix B is impossible. Any unit-time periodic L^4 estimate requires $\gamma \geq 1/4 - \alpha/8$, exactly the exponent already proved in the paper.

1 Source question

For $0 < \alpha < 1$, Thirouin [1] considers the periodic estimate

$$\|e^{-it|D|^\alpha} u\|_{L^4((0,1)_t \times \mathbb{T}_x)} \leq C\|u\|_{H^\gamma(\mathbb{T})}. \tag{1}$$

Appendix B proves (1) for

$$\gamma_* = \frac{1}{4} - \frac{\alpha}{8}$$

and asks whether the scaling candidate $\gamma_0 = \frac{1}{4} - \frac{\alpha}{4}$ might suffice. This improvement would lower a nonlinear well-posedness threshold from $\alpha > 2/3$ to $\alpha > 1/2$.

2 Sharp obstruction

A low-frequency Dirichlet kernel only gives the scaling exponent γ_0. The missing obstruction comes from a much wider packet centered at a large frequency. Since $\xi \mapsto \xi^\alpha$ has curvature of size $N^{\alpha-2}$ near N, approximately $N^{1-\alpha/2}$ consecutive modes remain coherent for the entire unit time interval after removing the affine phase.

Theorem 1. *Fix $0 < \alpha < 1$. If (1) holds for all trigonometric polynomials with a constant independent of u, then*

$$\gamma \geq \frac{1}{4} - \frac{\alpha}{8}.$$

Consequently (1) fails at $\gamma_0 = (1-\alpha)/4$. Combined with the source's upper bound, the sharp Sobolev exponent is $\gamma_* = 1/4 - \alpha/8$.

Proof. Fix a sufficiently small constant $0 < \delta < 1$ and, for a large integer N, set

$$M = \left\lfloor \delta N^{1-\alpha/2} \right\rfloor, \qquad u_N(x) = \sum_{j=0}^{M-1} e^{i(N+j)x}.$$

Because $M = o(N)$, all frequencies $N + j$ are comparable to N, and therefore

$$\|u_N\|_{H^\gamma} \asymp N^\gamma M^{1/2}. \tag{2}$$

B Some comments on the threshold $\alpha = \frac{2}{3}$

In this section, we discuss the bound $\alpha = \frac{2}{3}$ that naturally appears in the proof of theorem 3.

The crucial point, in the proof above, is to know which Strichartz estimate we are able to establish. In particular, following the strategy of [5], we would like to know for which values of the parameter γ the inequality

$$\|e^{-it|D|^\alpha} u(x)\|_{L^4((0,1)_t, L^4(\mathbb{T}_x))} \leq C\|u\|_{H^\gamma} \tag{33}$$

holds, whenever $u \in H^\gamma(\mathbb{T})$. If true, (33) would imply that equation (2) is well-posed in $H^{\alpha/2}$, provided that $2\gamma < \frac{\alpha}{2}$, and we could then adapt the arguments we developped in section 3.5 to prove that solutions are polynomially bounded.

Looking back at (20), and interpolating the $L^4((0,1)_t, L^\infty(\mathbb{T}_x))$ estimate we obtained with the trivial $L^\infty((0,1)_t, L^2(\mathbb{T}_x))$ one, we find a bound for $\|e^{-it|D|^\alpha} u(x)\|_{L^8((0,1)_t, L^4(\mathbb{T}_x))}$, so (33) is proved with $\gamma = \frac{1}{4} - \frac{\alpha}{8}$. The condition $2\gamma < \frac{\alpha}{2}$ exactly means that $\alpha > \frac{2}{3}$.

However, a scaling heuristic suggests that the natural value of γ should rather be $\gamma_0 := \frac{1}{4} - \frac{\alpha}{4}$. Here, the condition $2\gamma_0 < \frac{\alpha}{2}$ would enable us to extend the conclusions of theorem 3 until $\alpha = \frac{1}{2}$.

In [7], by a different method, Demirbas, Erdoğan and Tzirakis also prove a Strichartz estimate in the case $\alpha > 1$, but their result corresponds to ours (notice that in their work, they use other notations : what they call α is in fact half ours).

So far, we don't know if (33) can be proved with $\gamma = \gamma_0$. Usual counter-examples (such as localized functions) only confirm that the scaling exponent γ_0 is the best we can hope. Besides, the difficulty is not due to the particular framework of the torus, since when $\alpha < 1$ the speed of propagation of the waves (*i.e.* the group velocity) is finite.

Bibliography

Figure 1: The unit-time L^4 estimate and open question in Appendix B, page 27 of the source paper.

For $j \geq 0$, Taylor's theorem and the monotonicity of $|f''(r)| = \alpha(1-\alpha)r^{\alpha-2}$ give

$$\left|(N+j)^\alpha - N^\alpha - \alpha N^{\alpha-1} j\right| \leq \frac{\alpha(1-\alpha)}{2} N^{\alpha-2} j^2. \tag{3}$$

Write $v_N(t,x) = e^{-it|D|^\alpha} u_N(x)$. After discarding the unimodular factor $e^{iNx - itN^\alpha}$ and putting $y = x - \alpha N^{\alpha-1} t$, we obtain

$$|v_N(t,x)| = \left|\sum_{j=0}^{M-1} e^{i\{jy - t[(N+j)^\alpha - N^\alpha - \alpha N^{\alpha-1} j]\}}\right|.$$

Consider the moving tube

$$E_N = \{(t,x) \in [0,1] \times \mathbb{T} : |x - \alpha N^{\alpha-1} t|_{\mathbb{T}} \leq \delta/M\}.$$

For $(t,x) \in E_N$ and $0 \leq j < M$, the linear spatial phase has modulus at most δ, while (3) and the definition of M give

$$\left|t[(N+j)^\alpha - N^\alpha - \alpha N^{\alpha-1} j]\right| \leq \frac{\alpha(1-\alpha)}{2} \delta^2.$$

Choose δ so small that the total phase lies in $[-\pi/3, \pi/3]$. Every summand then has real part at least $1/2$, uniformly for all $(t,x) \in E_N$. Hence

$$|v_N(t,x)| \geq M/2 \quad ((t,x) \in E_N), \qquad |E_N| \asymp M^{-1}, \tag{4}$$

and thus

$$\|v_N\|_{L^4([0,1]\times\mathbb{T})} \gtrsim M^{3/4}. \tag{5}$$

If (1) held, (2) and (5) would imply

$$M^{3/4} \lesssim N^\gamma M^{1/2}, \qquad N^\gamma \gtrsim M^{1/4} \asymp N^{\frac{1}{4}-\frac{\alpha}{8}}.$$

Letting $N \to \infty$ forces $\gamma \geq 1/4 - \alpha/8$. Since

$$\left(\frac{1}{4} - \frac{\alpha}{8}\right) - \left(\frac{1}{4} - \frac{\alpha}{4}\right) = \frac{\alpha}{8} > 0,$$

the proposed exponent γ_0 fails; in fact, the quotient of the two sides of (1) diverges at least like $N^{\alpha/8}$ there. $\square$

3 Scope and checks

The time interval is the full unit interval: the packet stays coherent because the dispersion relation is nearly affine at high frequency, not because time is restricted. The moving spatial tube is harmless on the torus and has width comparable to M^{-1} for every time. All frequencies are positive, so $|D|^\alpha$ acts on the packet by $(N+j)^\alpha$ exactly.

This result answers only the precise linear question in Appendix B. It shows that estimate (1) cannot be the route for extending the paper's nonlinear result below $\alpha = 2/3$; it does not prove that the nonlinear threshold itself is sharp. The accompanying script checks the parameter exponents and a numerical phase bound, but the proof is exact.

A bounded search on 13 August 2026 covered the local corpus, exact question, both exponents, later periodic fractional Strichartz work, sharpness, and coherent-packet counterexamples. It found the question still present in the 2017 publication but no explicit resolution. This is not an exhaustive novelty claim. Human review should focus on the moving-tube phase estimate and the novelty status; it has not yet been completed.

References

[1] J. Thirouin, *On the Growth of Sobolev Norms of Solutions of the Fractional Defocusing NLS Equation on the Circle*, arXiv:1506.04181; Ann. Inst. H. Poincaré Anal. Non Linéaire 34 (2017), 509–531.